what the theory said?

Quantum physics needs an interpretation because it's not immediately clear what the theory is saying about the world. The mathematics of quantum physics is unfamiliar and abstruse, and the connection between that mathematics and the world we live in is hard to see. This is in stark contrast with the theory quantum physics replaced, the physics of Isaac Newton. Newton's physics describes a familiar and simple world with three dimensions, filled with solid objects that move in straight lines until something knocks them off their paths. The math of Newtonian physics specifies the location of an object using a set of three numbers, one for each dimension, known as a vector. If I'm on a ladder, two meters off the ground, and that ladder is three meters in front of you, then I could describe my position as (zero, three, two). The zero says that I'm not off to one side or the other, the three says I'm three meters in front of you, and the two says I'm two meters above you. It's fairly straightforward—nobody runs around deeply worried about how to interpret Newtonian physics.

But quantum physics is significantly stranger than Newtonian physics, and its math is stranger too. If you want to know where an electron is, you need more than three numbers—you need an infinity of them. Quantum physics uses infinite collections of numbers called *wave functions* to describe the world. These numbers are assigned to different locations: a number for every point in space. If you had an app on your phone that measured a single electron's wave function, the screen would just display a single number, the number assigned to the spot where your phone is. Where you're sitting right now, the Wave-Function-O-Meter™ might display the number 5. Half a block down the street, it'd display 0.02. That's what a wave function is, at its simplest: a set of numbers, fixed at different places.

Everything has a wave function in quantum physics: this book, the chair you're sitting in, even you. So do the atoms in the air around you, and the electrons and other particles inside those atoms. An object's wave function determines its behavior, and the behavior of an object's wave function is determined in turn by the Schrödinger equation, the central equation of quantum physics, discovered in 1925 by the Austrian physicist Erwin Schrödinger. The Schrödinger equation ensures that wave functions always change smoothly—the number that a wave function assigns to a particular location never hops instantly from 5 to 500. Instead, the numbers flow